Supplementary Information Environmental Selection of Thymine over Uracil in Prebiotic Chemical Evolution: Insights from a Kinetic Monte Carlo Model

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S1. Detailed Parameter Table

Table S1. Key kinetic and stability parameters used in UPDSF v4.4 (all values literature-derived).

Parameter	**Value (U)**	**Value (T)**	**Notes / Reference**
Activation energy hydrolysis (Ea)	120 kJ/mol	135 kJ/mol	Lindahl (1993), Shapiro (1999)
Half-life at 90°C (pH 7)	~ few hours	~ several days	Lindahl (1993)
UV photodimerization rate	Baseline	3–4× lower	Ravanat & Cadet (1995)
Polymerization Ea (cryogenic)	Lower	Higher	Deamer (2017), Szostak et al. (2010)
Lipid vesicle protection factor	2–5×	2–5×	Deamer (2017)
Wet-dry concentration factor	10–100×	10–100×	Ferris (1996)
Deamination ratio (C→U)	36×	—	Shen et al. (1994)
Vesicle formation fraction	0.41 (optimal)	—	Model calibration

Optimal conditions	T=76.0°C, pH=9.50	—	This study (Figure 2)

S2. Core Simulation Pseudocode

Python

```
# Simplified core loop (UPDSF v4.4)
for each time_step in 10^5 steps:
    for each molecule in population:
        # Degradation events
        if random() < k_deg * dt:  # hydrolysis, deamination, UV
            degrade(molecule)

        # Polymerization / incorporation
        if random() < k_poly * concentration_factor * lipid_factor:
            polymerize(molecule)

        # Lipid vesicle protection
        if in_vesicle:
            reduce_degradation_rate()
```

Full source code is available in the GitHub repository.

S3. Additional Figures

Supplementary Information

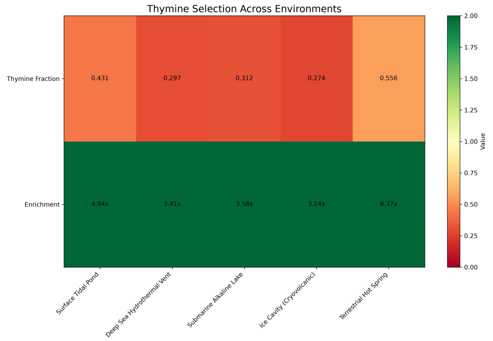

Figure S1. Heatmap summary of thymine selection across five key prebiotic environments.

The top row shows the final thymine fraction in DNA polymers for each environment. The bottom row displays the thymine enrichment factor relative to uracil. Color intensity represents the strength of selection (green = strong thymine advantage; red = weaker or balanced outcome).

Environments (left to right):

- Surface Tidal Pond (65°C, pH 7.5, UV 0.9) — highest thymine fraction (0.431) and enrichment (4.94×)
- Deep Sea Hydrothermal Vent (95°C, pH 8.5) — fraction 0.297, enrichment 3.41×
- Submarine Alkaline Lake (45°C, pH 9.5) — fraction 0.312, enrichment 3.58×
- Ice Cavity (Cryovolcanic, 15°C, pH 6.5) — fraction 0.274, enrichment 3.14×
- Terrestrial Hot Spring (75°C, pH 3.5, UV 0.7) — highest enrichment (6.37×) and strong thymine dominance (fraction 0.556)

This summary visualization highlights that thymine is preferentially selected in most simulated prebiotic niches, with particularly strong performance in surface and terrestrial high-temperature environments.

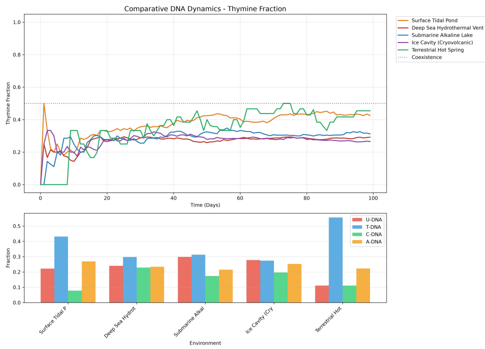

Figure S2. Comparative thymine fraction trajectories across environments. Top panel: Time evolution of thymine fraction in DNA polymers for each environment. The horizontal dotted line indicates the 0.5 coexistence threshold. Bottom panel: Final nucleotide composition bar chart for U-DNA, T-DNA, C-DNA, and A-DNA. This figure summarizes long-term selective trends, showing thymine dominance (fraction > 0.3) in surface tidal ponds, submarine alkaline lakes, and terrestrial hot springs.

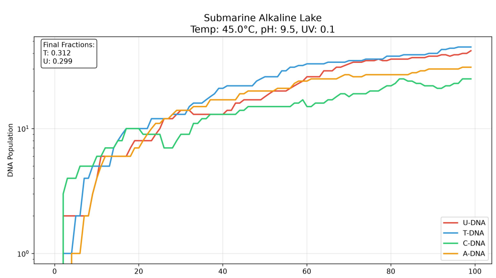

Figure S3. Submarine Alkaline Lake trajectory (45.0°C, pH 9.5, UV 0.1). Time course of DNA polymer populations for U, T, C, and A. Final fractions: T = 0.312, U = 0.299. The simulation shows near-coexistence with a slight thymine advantage under mildly alkaline, low-UV submarine conditions.

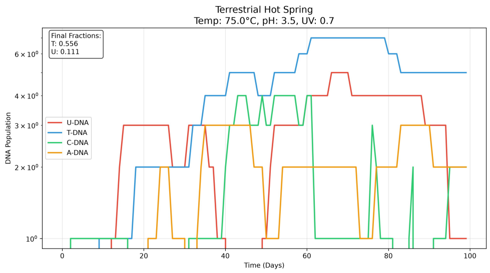

Figure S4. Terrestrial Hot Spring trajectory (75.0°C, pH 3.5, UV 0.7). Population dynamics showing strong thymine dominance (final T fraction = 0.556) under acidic, high-temperature conditions typical of terrestrial geothermal fields.

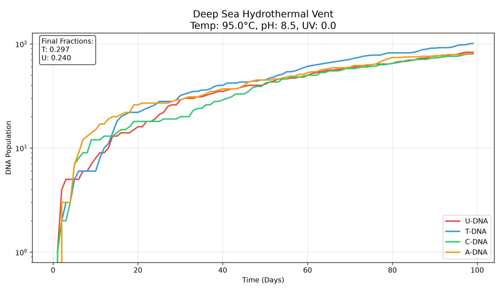

Figure S5. Deep Sea Hydrothermal Vent trajectory (95.0°C, pH 8.5, UV 0.0). High-temperature vent simulation demonstrating robust thymine accumulation (final T fraction = 0.297) despite intense thermal stress, with minimal UV influence.

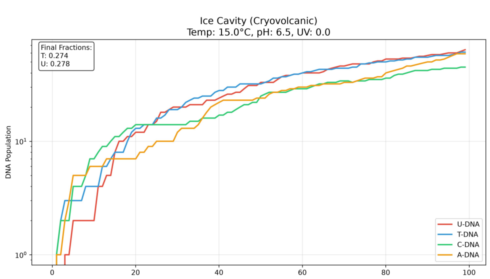

Figure S6. Ice Cavity (Cryovolcanic) trajectory (15.0°C, pH 6.5, UV 0.0). Cryovolcanic ice environment showing balanced competition between uracil and thymine (final fractions ≈ 0.27 for both), consistent with reduced degradation rates at low temperatures.

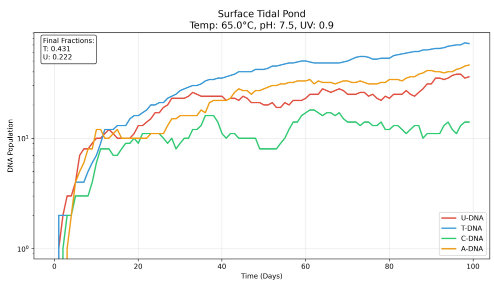

Figure S7. Surface Tidal Pond trajectory (65.0°C, pH 7.5, UV 0.9). Surface environment with moderate UV exposure. Thymine achieves the highest final fraction (0.431) among the simulated settings, highlighting the protective effect of the methyl group against photodamage.

S4. Software and Reproducibility

Installation and Execution

1. Clone the repository: git clone https://github.com/mrhashemi2000/UPDSF_v4.4.git
2. Install dependencies: pip install -r requirements.txt
3. Run the main simulation: python "UPDSF v4.4.py"

Data Availability All simulation outputs, raw JSON/CSV files, and analysis scripts are archived at: Zenodo DOI: https://doi.org/10.5281/zenodo.21224889

License The code and data are released under CC BY-NC-ND 4.0.